

Channu Praveen Kumar — Resume

Python Full Stack & AI Engineer | FastAPI · Angular · LLMs · Voice AI | 3+ Years Experience

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Professional Summary

Python Full Stack Engineer and AI Engineer with 3+ years of experience building scalable web applications, multi-tenant SaaS platforms and production LLM features using Python, FastAPI, Django, PostgreSQL and Angular.

Founder and lead engineer of AiOpsCare, a multi-tenant hospital operations SaaS with schema-per-tenant PostgreSQL isolation, a multilingual voice assistant (speech-to-text, LLM, text-to-speech) and LLM-powered chat and reports, deployed on Render, Neon and Netlify.

Strong in REST API design, multi-tenant backend architecture, LLM integration (OpenAI, Groq, Llama 3.3), prompt engineering, structured LLM outputs, debugging production issues and performance optimisation. Builds end-to-end features independently, from database schema to UI.

Technical Skills

- **Languages:** Python, TypeScript, JavaScript, SQL
- **Backend:** FastAPI, Django, Django REST Framework, SQLAlchemy, Alembic, Pydantic, Celery, asyncio, Clean Architecture
- **AI / LLM:** OpenAI (GPT-4o, GPT-4o-mini), Groq (Llama 3.3 70B), Anthropic Claude, Gemini, Ollama (local LLMs), prompt engineering, structured JSON outputs, multi-provider LLM routing and fallback, tenant-scoped context injection, AI chat over business data
- **Voice AI:** speech-to-text, text-to-speech (TTS), multilingual AI (English, Hindi, Telugu, Tamil), Indic transliteration, Web Speech API
- **Frontend:** Angular 18, Signals, RxJS, PrimeNG, ngx-translate (i18n), HTML5, CSS3, Bootstrap, responsive UI

- **APIs & Security:** REST APIs, OpenAPI/Swagger, Postman, JWT, OAuth2 PKCE, RBAC, WebSockets
- **Databases:** PostgreSQL (schema-per-tenant multi-tenancy), MySQL, SQL Server, schema design, stored procedures, query optimisation
- **Cloud & DevOps:** Docker, AWS EC2, Render, Neon, Netlify, Git, GitHub, Linux

Professional Experience

Founder & Lead Engineer (Python Full Stack / AI) — AiOpsCare, Hyderabad | Oct 2025 – Present

- Built a multi-tenant hospital operations SaaS from scratch with Python, FastAPI, PostgreSQL and Angular 18, using a Clean Architecture backend (routers, services, repositories).
- Designed schema-per-tenant PostgreSQL isolation with tenant context resolved from JWT, so every request is routed to the correct hospital's schema.
- Implemented JWT authentication and role-based access control for Super Admin, Hospital Admin, Supervisor and Staff roles.
- Delivered modules for tickets (SLA enforcement, automatic worker assignment), assets, preventive maintenance, contracts (AMC/CMC), checklists, housekeeping, laundry, utility meters, incidents/CAPA and approval workflows.
- Built a multilingual voice assistant (English, Hindi, Telugu, Tamil) combining speech-to-text, LLM reasoning and text-to-speech for hands-free ticket creation from voice notes.
- Built a provider-agnostic LLM client supporting OpenAI (GPT-4o) and Groq (Llama 3.3 70B) with automatic fallback and structured JSON outputs, so a provider outage does not stop operations.
- Built an AI dashboard assistant and narrative reports with tenant-scoped context injection, so natural-language queries resolve only against the requesting hospital's data.
- Improved TTS pronunciation for Indian languages with a transliteration layer and added an LRU cache for repeated phrases.
- Fixed mobile voice issues (Android microphone conflicts, blocked audio autoplay) with session-scoped audio streams, retries and timeouts.
- Deployed backend on Render with Neon PostgreSQL and frontends on Netlify; fixed schema drift across tenant schemas with idempotent SQL migrations.

Full Stack Developer — Nushift Technologies, Hyderabad | Jul 2024 – Sep 2025

- Developed Angular modules for a healthcare platform with real-time data features.
- Integrated REST APIs and optimised data loading, improving page performance and responsiveness.
- Debugged and fixed production issues, improving application stability.
- Built reusable UI components shared across modules.
- Worked with backend teams to improve API response efficiency in agile sprints.

Full Stack Software Developer — SNAD Developers Pvt Ltd, Hyderabad | Apr 2022 – Feb 2024

- Built REST APIs and backend services and integrated them with Angular frontends for an HRMS product.
- Developed responsive UI using Angular, HTML, CSS and Bootstrap.
- Optimised SQL queries and indexes, improving performance on data-heavy screens.
- Fixed production bugs and used Git for version control and team collaboration.

Projects

SwarmAI — Distributed Local LLM Inference (Open Source) | Python, FastAPI, asyncio, Ollama, AWS EC2

- Built a coordinator-worker system that distributes LLM workloads across machines running local Ollama models.
- Implemented heartbeat health checks, least-busy worker routing and retry logic for fault tolerance.
- Measured a 1.66x speedup on two nodes versus one (113.3s to 68.3s) in a benchmark test environment.
- Built a Python CLI for joining workers and running workflows; tested internet-accessible nodes via AWS EC2 and ngrok.

DevSparkAI Social Hub — AI Social Media Management SaaS | Angular, RxJS, FastAPI, PostgreSQL, LLM APIs | [Live](#)

- Built AI content planning: users give a topic and timeframe and receive a structured content plan with platform-specific post variants.
- Added a provider layer for multiple LLMs (OpenAI, Gemini and others) with bring-your-own encrypted API keys.

- Implemented JWT authentication, X/Twitter OAuth2 PKCE publishing, post scheduling and analytics.

HRMS Portal — Employee Management System | Python, Django, Django REST Framework, PostgreSQL, Angular

- Developed employee management with authentication, role-based permissions and DRF serializer validation.
- Improved report performance by optimising Django ORM queries and database indexes.

Education

- **B.Sc Electronics** — Bhavans Vivekananda College, Hyderabad | 2022
- **Intermediate (MPC)** — Siva Shivani Junior College | 2019